KUKU — AI GENERALIST ASSIGNMENT

Problem 1: Find the Promos (Ads) | Candidate: Srikrishna Valepe | AI Generalist Solution

Live Prototype URL: https://kuku-promo-detector.onrender.com | **GitHub Repo:** https://github.com/srikrishna6174/kuku-ai-promo-detector
Status: Deployed & Live on Render Cloud • 20/20 Passing Automated Unit Tests • Multi-Format Universal Parser • Multi-LLM Failover

1. Problem Choice, Framing & How the Project Works

Why Problem 1? Microdramas are a high-velocity vertical video format where editors manually scrub 50+ episode timelines to find 20–60s hooks for TikTok, IG Reels, and YouTube Shorts. Our production tool automates promo scouting from raw screenplays into exact timestamp cut sheets with zero hallucination.

How It Works End-to-End:

1. Ingestion: Universal parser accepts `.txt`, `.srt`, `.vtt`, or zip archives of raw screenplays, normalizing dialogue cues, character names, and millisecond timestamps.

2. Deterministic Candidate Windows: Sliding window generator extracts valid scene boundaries conforming strictly to 20s–60s duration constraints with +0.75s post-dialogue visual reaction buffer.

3. Multi-Tier Detection Engine: Evaluates clips through 3 modes: (a) *Deterministic Rules* (0–3s hook disruption, conflict escalation, final cliffhanger peak); (b) *Generative AI* (semantic tension, dialogue punchline, 'Why it works' rationale); (c) *Hybrid* (LLM-as-a-Judge validation + auto failover between Gemini & OpenAI).

4. Editor Studio: Editors review collapsible promo categories, adjust/pin custom manual timestamp cuts, and export clean cut sheets, Premiere/FCP marker CSVs, and lossless FFmpeg stream-copy scripts.

2. Architectural Division: Deterministic Rules vs Generative AI

Component	Deterministic Code / Simple Rules	Generative AI / LLM Layer
Responsibilities	• Screenplay parsing (.txt, .srt, .vtt, .zip) & timestamp extraction. • Sliding window generation (20s–60s bounds, +0.75s reaction buffer). • Anchor detection: Dialogue starts, high-intensity tones, confrontation verbs. • Mathematical scoring of 0–3s hook & final cliffhanger.	• Semantic tension assessment, hook/cliffhanger summaries & 'Why it works'. • LLM-as-a-Judge pass & auto failover on outage/rate limit. • Swappable provider adapter (Gemini & OpenAI; pluggable REST APIs).
Throughput & Scope	Offline-capable on CPU; scans 50 episodes in <2.0s with zero API cost.	Evaluates filtered top candidate windows (~2–4/ep) or user-chosen episodes.

3. Scoring Rubric & Window Selection Logic

Candidate clips are evaluated on a composite 100-point scale across three weighted dimensions:

• **Hook Score (40%):** Evaluates the first 3.0 seconds. Looks for high-stakes disruption verbs (*'marry', 'turn around', 'betray', 'enough'*), confrontational tones (*Venomous, Disgusted, Screaming*), or physical shock actions (*slap, tear, doors burst*).

• **Cliffhanger Score (35%):** Evaluates the final 3.0–5.0 seconds. Looks for unresolved interrogatives (*'?'*), interrupted statements (*'...'*), acoustic swells (*'ominous cliffhanger peak'*), or visual freeze/cut-to-black.

• **Standalone Coherence (25%):** Requires >= 2 distinct speakers in conflict and 4–10 dialogue turns. Penalizes slow monologue exposition.

4. Strategic Campaign Rollout: Which Promos to Post First & Why?

Ranking microdrama promos is not just about raw emotional score; it requires a **funnel-aligned publishing strategy**:

Tier & Priority	Target Episodes	Strategic Rationale & Placement
Tier 1: Post First (Cold User Acquisition)	**Episodes 1 – 3** (Rank #1: Ep 3, 00:25–00:56 Rank #2: Ep 1, 01:15–01:38)	**Zero-Lore Universality:** The wedding betrayal, sister stealing the groom, and mysterious hero's intervention require zero prior knowledge. Highest conversion from cold scroller to initial app install.
Tier 2: Post Second (Viral Escalation & Shares)	**Episodes 4 – 15** (Identity reveals & public confrontation)	**Power Reversal & Catharsis:** Juliana standing her ground and the Duke's true status emerging. Drives high comment velocity and algorithmic engagement on TikTok/Instagram.
Tier 3: Post Third (Retargeting & Churn Prevention)	**Episodes 16 – 50** (High-cliffhanger mid-season)	**Binge Triggers:** Life-or-death and high-stakes ultimatums targeted at warm audiences who installed the app but stalled at Episode 5.

5. Honest Quality Evaluation & Failure Modes ('Where does the tool pick badly?')

Automated audit (*src/quality_checker.py*) across all extracted promos scored 26 clean and flagged 74 edge cases:

• **1. Premature Cut Buffer (58% of clips):** Scripts end dialogue abruptly (e.g. 00:32.40). Cutting there clips visual reaction / audio sting. *Mitigation:* Parser accounts for +0.75s reaction buffer.

• **2. Off-Target Secondary Characters (32% of clips):** Shouting matches between minor cast score high lexically but fail as cold UA ads. *Mitigation:* Lead-character weighting bonus.

• **3. Lore & Exposition Overload:** High-word-count scenes (>25 words/turn) bore scrollers. *Mitigation:* Turn density penalizer.

• **4. Visual-Only Ambiguity:** Silent action beats (slaps, chases) are underestimated without multimodal vision.

6. AI Tools Used & One Place AI Got It Wrong

• **Tools Used:** Antigravity coding agent for full pipeline and Streamlit studio development; Gemini & OpenAI REST APIs for semantic analysis, tension detection, and LLM-as-a-Judge validation.

• **Where AI Got It Wrong:** Direct LLM prompting on screenplays caused 3 critical flaws: (1) Cutting on the exact final word, truncating audio tails; (2) Drifting beyond the 20–60s bounds; (3) Hallucinating timestamps not matching subtitle sync. Wrapping the LLMs inside deterministic window math, strict duration guards, and 20 automated unit tests eliminated these issues completely.

7. Production Workflow, Scaling & Deliverables (Moments Only — No Overlays/Editing)

• **Scaling & Human-in-the-Loop Workflow:** Rule filtering scans 1,000+ episodes in <45s on standard CPU, dispatching only top candidates to LLMs. Editors review ranked 20–60s cuts, pin manual adjustments, and export cut sheets without baked video overlays or artificial music.

• **Deliverables:** (1) **Live Cloud App:** https://kuku-promo-detector.onrender.com; (2) **GitHub Repo:** https://github.com/srikrishna6174/kuku-ai-promo-detector; (3) **Outputs Pack:** Structured *outputs/promos/* (JSON, CSV, report, cut sheets), *outputs/quality_audit/*, and *outputs/editor_exports/*.